NAME: ASHOK

REG.NO: 192312558

Ex 2: 8-Queen Problem

CODE:

```
    def is_safe(board, row, col):
  for i in range(row):
    if board[i] == col or abs(board[i] - col) == abs(i - row):
      return False
  return True
def solve_n_queens(row, board, n):
  if row == n:
    return True
  for col in range(n):
    if is_safe(board, row, col):
      board[row] = col
      if solve_n_queens(row + 1, board, n):
        return True
  return False
n = 8
board = [-1] * n
solve_n_queens(0, board, n)
for r in range(n):
  line = ["." for _ in range(n)]
  line[board[r]] = "Q"
```

```
    print(" ".join(line))
```

OUTPUT:

